The Hilbert-Space Quadrature Conjecture: Time as the Wave-Function-Generated Quadrature of Circular Phase

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We develop a geometric and operational reading of relational quantum time, which we call the *Hilbert-space quadrature conjecture*: physical time is not an external parameter but is generated by the universal wave function through the quadratic (Born) map acting on the circular $U(1)$ phase structure of a clock sector. Technically the framework rests on the Page–Wootters conditional-probability construction with shift-covariant clock POVMs, as developed in Refs. [10–12]; our contributions are interpretive and clarifying rather than a new dynamics. (i) Using the Zak (Bloch–Floquet) transform we make precise how universal-cover clocks ($\mathbb{R}$) relate to cyclic clocks (S^1): exact constraint solutions for arbitrary system spectra live in twisted circle sectors, and a cyclic clock read without its winding number preserves coherence only between levels whose energy difference is an integer multiple of the clock quantum $\hbar\omega$. (ii) We separate the raw clock marginal, the joint clock–system quadrature and the conditional system state, recovering the uniform marginal and exact conditional Schrödinger evolution for ideal clocks. (iii) We distinguish two finite-clock effects that are frequently conflated: unsharp clock readout, which produces a *time-independent* Gaussian suppression of energy coherences, and diffusive clock error, which—under an explicitly stated model assumption—yields the double-commutator Gorini–Kossakowski–Sudarshan–Lindblad equation. (iv) We examine a circuit-QED test and show that its predicted signature hinges on which energy gap enters the rate; the most natural identification is already excluded by measured qubit coherence times. Finally we outline relativistic, gravitational and cosmological extensions, separating established results from heuristic proposals.

I. INTRODUCTION

In non-relativistic quantum mechanics time $t \in \mathbb{R}$ enters the Schrödinger equation as a classical parameter external to the Hilbert space $\mathcal{H}$. Pauli observed [1] that a Hamiltonian bounded below admits no self-adjoint time operator $\widehat{T}$ with $[\widehat{T}, \widehat{H}] = i\hbar$. In canonical quantum gravity and other reparametrization-invariant theories the Wheeler–DeWitt constraint $\widehat{C}|\Psi\rangle = 0$ removes external time altogether [2, 3].

Two programs recover temporal order from a stationary state: the **Page–Wootters (PW) mechanism** [4, 5], which extracts dynamics from entanglement between a clock $\mathcal{H}_C$ and a system $\mathcal{H}_S$, and the **thermal-time hypothesis** of Connes and Rovelli [6], which identifies time flow with the Tomita–Takesaki modular group of a state.

Our starting point is the geometry of complex Hilbert space. Normalized vectors form the sphere $S(\mathcal{H})$; physical rays form $\mathbb{P}(\mathcal{H}) = S(\mathcal{H})/U(1)$ [7, 8], giving the principal circle bundle

$$U(1) \hookrightarrow S(\mathcal{H}) \xrightarrow{\pi} \mathbb{P}(\mathcal{H}). \tag{1}$$

The global phase along a fibre is unobservable. When a universal state $|\Psi\rangle \in \mathcal{H}_C \otimes \mathcal{H}_S$ entangles a clock with a system, however, the *relative* phases of the clock's energy components become a clock reading, and the quadratic Born map—together with its phase-space form, the Wigner transform [9]—turns ("squares") these circular amplitudes into a nonnegative distribution over clock readings and conditional system states. The slogan of this paper is that *the wave function of the clock sector, passed through the quadratic map, is what turns the circular phase geometry of Hilbert space into observable temporal order.*

Relation to prior work. The mathematical ingredients are established. PW conditioning with shift-covariant clock POVMs, including clocks whose Hamiltonians are bounded below and clocks interacting with the system, was developed by Smith and Ahmadi [10]; the equivalence of the PW, relational-Dirac-observable and reduced-phase-space pictures, with covariant POVMs for non-ideal clocks, was established by Höhn, Smith and Lock [11]; a rigorous relative-time version is due to Loveridge and Miyadera [12]. Covariant phase observables go back to Holevo [13] and are reviewed in Ref. [14]. Loss of coherence due to imperfect clocks was analyzed by Egusquiza, Garay and Raya [15], by Milburn [16] and by Gambini, Porto and Pullin [17]. Relativistic quantum clocks and gravitational time dilation in this language are treated in Refs. [18–21].

What is new here. (a) The organizing interpretation (the quadrature reading of Eq. (1)) and the conjecture built on it (Conjecture 1). (b) A precise line-versus-circle statement via the Zak transform (Proposition 1) and the *winding-marginalization* result (Proposition 2), which shows what a cyclic clock can and cannot time. (c) A clean separation of unsharp readout from diffusive clock error (Sec. V), making explicit the assumption behind the Lindblad form.

(d) A critical assessment of a circuit-QED signature, including the regime in which it could be tested (Sec. VI). The relativistic and cosmological sections (Secs. VIII–IX) collect known results in the present language and flag which statements are heuristic.

What is not claimed. We do not claim to solve the problem of time, to derive the Hawking temperature, or to provide predictions that differ from standard quantum mechanics with a quantum clock. Without an additional law fixing the clock-error parameter T_C of Sec. V, the conjecture is an interpretation that is empirically equivalent to the PW formalism.

Plan of the paper. Sections II–IV build the core: the line-versus-circle kinematics, the covariant clock POVM, and the tri-fold quadrature with its phase-space form. Section V treats finite clocks, and Sec. VI immediately confronts the resulting master equation with a laboratory setting. Section VII relates conditional time to modular (thermal) time. Sections VIII and IX then apply the framework rung by rung: flat spacetime, static gravity, the horizon, and finally quantum cosmology, where no background clock exists.

II. KINEMATICS: THE LINE AND THE CIRCLE

A. The spectral obstruction on a compact clock

Let the clock live on the circle, $\mathcal{H}_C = L^2(S^1, d\theta/2\pi)$ with $\widehat{N}|n\rangle = n|n\rangle$, $n \in \mathbb{Z}$, and impose

$$\widehat{C}_{\rm circ}|\Psi\rangle = \left(\hbar\omega \widehat{N} \otimes I_S - I_C \otimes \widehat{H}_S\right)|\Psi\rangle = 0. \tag{2}$$

In the product eigenbasis $|n\rangle|E_k\rangle$ this requires $\hbar\omega n = E_k$: a nonzero normalizable solution supported on a level E_k exists only if $E_k/\hbar\omega \in \mathbb{Z}$. Equivalently, the conditional vector $|\Psi(\theta)\rangle = e^{-i\widehat{H}_S\theta/\hbar\omega}|\Psi(0)\rangle$ is single-valued on S^1 only if $e^{-2\pi i\widehat{H}_S/\hbar\omega} = I_S$ on its support. For a generic Hamiltonian (incommensurate gaps or continuous spectrum) $\ker \widehat{C}_{\rm circ} = \{0\}$.

B. Universal cover and group averaging

On the universal cover $\mathcal{H}_C^{\rm univ} = L^2(\mathbb{R}, dt)$, with $\langle t|t'\rangle = \delta(t-t')$ and $\widehat{P}_t = -i\hbar\, d/dt$, the constraint

$$\widehat{C}_{\rm univ} = \widehat{P}_t \otimes I_S + I_C \otimes \widehat{H}_S \tag{3}$$

has 0 in its continuous spectrum. Physical states are distributions in a Gel'fand triple $\Phi \subset \mathcal{H}^{\rm kin} \subset \Phi^\times$ obtained by group averaging (refined algebraic quantization) [22],

$$\mathbb{P}_{\rm phys} = \frac{1}{2\pi\hbar}\int_{-\infty}^{\infty} ds\, e^{is\widehat{C}_{\rm univ}/\hbar}, \qquad \langle\phi|\psi\rangle_{\rm phys} = \langle\phi|\mathbb{P}_{\rm phys}|\psi\rangle_{\rm kin}, \tag{4}$$

and satisfy $\langle t|\Psi\rangle = e^{-i\widehat{H}_S t/\hbar}|\Psi(0)\rangle$ for any $\mathrm{Spec}(\widehat{H}_S) \subset \mathbb{R}$. The price is that the ideal clock momentum $\widehat{P}_t$ is unbounded in both directions, so this clock has no ground state; this is precisely the situation Pauli's argument forbids for a physical clock, and it motivates the semi-bounded clocks of Sec. III.

C. Zak decomposition of the line into twisted circles

To relate the line and the circle we use the Zak (Bloch–Floquet) transform [23]. Let $T_0 = 2\pi/\omega$.

Definition 1 (Zak transform)**.** For $f \in \mathcal{S}(\mathbb{R})$ define

$$(\mathcal{Z}f)(\theta,\kappa) = \sum_{m\in\mathbb{Z}} e^{-i\kappa m}\, f\!\left(\frac{\theta+2\pi m}{\omega}\right), \qquad \theta\in\mathbb{R},\ \kappa\in[0,2\pi). \tag{5}$$

Proposition 1 (Line = direct integral of twisted circles)**.** *(i) $\mathcal{Z}$ extends to a unitary map from $L^2(\mathbb{R},dt)$ onto $\int^{\oplus}_{[0,2\pi)} \mathcal{H}_\kappa \frac{d\kappa}{2\pi}$, where $\mathcal{H}_\kappa$ is the space of functions on $[0,2\pi)$ obeying the twisted boundary condition $g(\theta+2\pi) = e^{i\kappa}g(\theta)$, with norm $\|f\|^2 = \omega^{-1}\int_0^{2\pi}\frac{d\kappa}{2\pi}\int_0^{2\pi} d\theta\,|(\mathcal{Z}f)(\theta,\kappa)|^2$. (ii) Translations are intertwined with rotations: $(\mathcal{Z}\,T_\tau f)(\theta,\kappa) = (\mathcal{Z}f)(\theta-\omega\tau,\kappa)$ with $(T_\tau f)(t) = f(t-\tau)$. (iii) $\widehat{P}_t$ acts fibrewise as $-i\hbar\omega\,\partial_\theta$, whose spectrum on $\mathcal{H}_\kappa$ is $\hbar\omega\,(\mathbb{Z}+\kappa/2\pi)$. (iv) The periodization map $(\Pi_\omega f)(\theta) = \sum_m f((\theta+2\pi m)/\omega)$ is the $\kappa = 0$ fibre of $\mathcal{Z}$.*

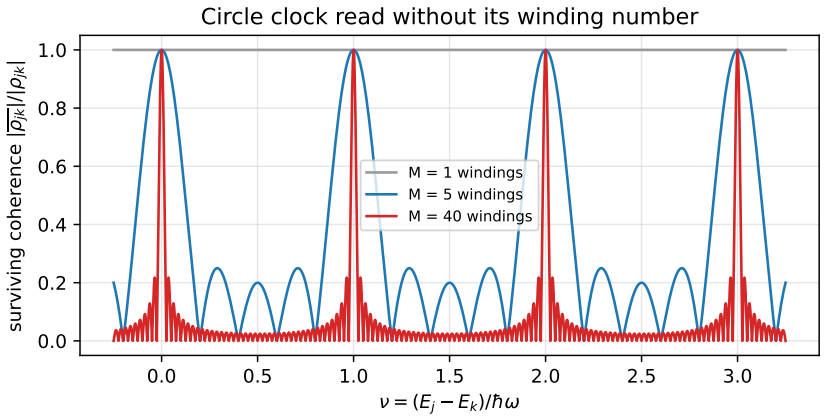


FIG. 1. Coherence surviving winding marginalization, $|D_M(\nu)|$ of Eq. (6), as a function of $\nu = (E_j - E_k)/\hbar\omega$. As the number of uncounted windings M grows, only commensurate energy differences survive.

Proof. Quasi-periodicity follows by shifting $m \to m+1$ in Eq. (5). For (i), integrating $|\mathcal{Z}f|^2$ over κ kills the cross terms $m \neq m'$, leaving $\sum_m |f((\theta+2\pi m)/\omega)|^2$, whose integral over one period is $\omega\|f\|^2$; surjectivity follows from the inverse formula $f(t) = \int_0^{2\pi} \frac{d\kappa}{2\pi}(\mathcal{Z}f)(\omega t, \kappa)$. (ii) is immediate, (iii) follows by differentiating under the sum with $\partial_\theta = \omega^{-1}\partial_t$ and noting that $e^{i\nu\theta} \in \mathcal{H}_\kappa$ iff $\nu \in \mathbb{Z} + \kappa/2\pi$, and (iv) is the definition at $\kappa = 0$. □

Corollary 1 (Where physical states live)**.** *The generalized physical state* $\langle t|\Psi_E\rangle = e^{-iEt/\hbar}\psi_E$ *has Zak transform* $e^{-iE\theta/\hbar\omega}\, 2\pi \sum_{l\in\mathbb{Z}} \delta(\kappa + 2\pi E/\hbar\omega - 2\pi l)\, \psi_E$*: it is supported in the single fibre* $\kappa_E \equiv -2\pi E/\hbar\omega \pmod{2\pi}$*, where it is an eigenfunction of the fibre clock generator. In particular the untwisted circle (*$\kappa = 0$*, i.e. periodization) contains only commensurate energies* $E \in \hbar\omega\mathbb{Z}$*.*

Thus periodization on its own does *not* remove the obstruction of Sec. II.A (Poisson summation reinstates it); the full family of twisted sectors does. A cyclic clock with a fixed twist κ (an Aharonov–Bohm-like flux through the clock circle) can time exactly those system energies with $E/\hbar\omega \equiv -\kappa/2\pi \pmod 1$.

D. Reading a cyclic clock without its winding number

Operationally a cyclic clock records an angle $\theta = \omega t \bmod 2\pi$ but not the winding number $m = \lfloor \omega t/2\pi \rfloor$. The following result quantifies the cost.

Proposition 2 (Winding marginalization)**.** *Let the system evolve as* $\rho_S(t) = U(t)\rho_0 U^\dagger(t)$*,* $U(t) = e^{-i\widehat{H}_S t/\hbar}$*, and suppose the clock reading* $\theta \in [0, 2\pi)$ *is recorded while the winding number* $m \in \{0, \dots, M-1\}$ *is not (uniform prior). Then the conditional state is*

$$\bar{\rho}_S(\theta) = \frac{1}{M}\sum_{m=0}^{M-1} U(\theta/\omega + mT_0)\,\rho_0\, U^\dagger(\theta/\omega + mT_0), \qquad \bar{\rho}_{jk}(\theta) = \rho_{jk}(\theta/\omega)\, D_M(\nu_{jk}), \tag{6}$$

with $\nu_{jk} = (E_j - E_k)/\hbar\omega$ *and* $D_M(\nu) = \frac{1}{M}\sum_{m=0}^{M-1} e^{-2\pi i m\nu}$*,* $|D_M(\nu)| = \left|\frac{\sin(\pi M\nu)}{M\sin(\pi\nu)}\right|$*. Hence* $|D_M| = 1$ *for* $\nu_{jk} \in \mathbb{Z}$*, while for* $\nu_{jk} \notin \mathbb{Z}$*,* $|D_M| \le (M|\sin\pi\nu_{jk}|)^{-1} \to 0$*.*

Proof. Bayes' rule with a uniform prior over the M preimages $t_m = \theta/\omega + mT_0$ gives the mixture. In the energy basis $U(t_m)$ contributes $e^{-i(E_j-E_k)t_m/\hbar} = e^{-i\omega_{jk}\theta/\omega}e^{-2\pi i m\nu_{jk}}$; summing the geometric series gives D_M. □

A cyclic clock is therefore a faithful timekeeper over many cycles only for commensurate spectra or when the winding number is counted, in which case the pair (θ, m) is equivalent to the universal cover. In the language of the title, the circle becomes a line when the turns are counted (Fig. 1). It is this bookkeeping of windings, not interference, that straightens the circle; interference among clock levels does a different job, localizing a moment on the line (Proposition 4).

E. Operational compactifications

When one works directly on a compact clock, incommensurability is handled either by (1) a *spectral window*: normalizable states $|\Psi_\epsilon\rangle \propto f(\widehat{C}/\epsilon)|\eta\rangle$ with f peaked at zero, satisfying $\langle\widehat{C}^2\rangle \lesssim \epsilon^2$; or (2) a finite Pegg–Barnett clock [24] $\mathcal{H}_C^{(N)} = \mathbb{C}^N$ with orthonormal phase states

$$|\theta_k^{(N)}\rangle = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} e^{in\theta_k}|n\rangle, \qquad \theta_k = \frac{2\pi k}{N}. \tag{7}$$

Remark 1 (Sign conventions and semi-bounded clocks). The overall sign of a constraint is immaterial, but the *relative* sign of clock and system energies is not. In Eq. (2) the clock enters with the opposite sign to the system, so that a clock with $\mathrm{Spec}(\widehat{N}) = \mathbb{N}_0$ pairs with system energies $E_k = \hbar\omega n_k \geq 0$ and conditioning on the bra $\langle\theta|$ of Sec. III gives forward evolution. Because a semi-bounded clock covers only part of the real line, only system energies in a corresponding window (after a constant shift of $\widehat{H}_S$) are represented. The Hardy restriction below is therefore a *selection rule*, and this is the mechanism behind the positive-frequency selection of Sec. VIII A.

III. THE COVARIANT CLOCK POVM

Let the physical clock be the Hardy space $\mathcal{H}_C = H^2(S^1)$ spanned by $\{|n\rangle\}_{n\geq 0}$ with $\widehat{H}_C = \hbar\omega\widehat{N}$, $\mathrm{Spec}(\widehat{N}) = \mathbb{N}_0$. Following Holevo [13] define the generalized vectors

$$|\theta) = \sum_{n=0}^{\infty} e^{in\theta}|n\rangle, \qquad \theta \in [0, 2\pi). \tag{8}$$

Because the sum starts at $n = 0$ these vectors are not orthogonal. Their overlap is the boundary value of the Szegő kernel,

$$(\theta|\theta') = \sum_{n=0}^{\infty} e^{in(\theta'-\theta)} = \frac{1}{1 - e^{i(\theta'-\theta+i0^+)}} = \pi\,\delta_{2\pi}(\theta'-\theta) + \frac{1}{2} + \frac{i}{2}\,\mathrm{P.V.}\cot\frac{\theta'-\theta}{2}, \tag{9}$$

where the delta function is produced by the $+i0^+$ prescription (Sokhotski–Plemelj) and must not be added a second time.

Proposition 3 (Covariant circular clock POVM). *$E(\Delta) = \int_\Delta E(d\theta)$, $E(d\theta) = \frac{1}{2\pi}|\theta)(\theta|\,d\theta$, defines a POVM on the Borel sets of S^1 with: (1) positivity, $\langle\chi|E(\Delta)|\chi\rangle = \int_\Delta \frac{d\theta}{2\pi}|(\theta|\chi\rangle|^2 \geq 0$; (2) completeness, $E(S^1) = I_C$; (3) covariance, $U(\phi)E(\Delta)U^\dagger(\phi) = E(\Delta\oplus\phi)$ with $U(\phi) = e^{i\phi\widehat{N}}$; (4) the Pegg–Barnett PVMs $E^{(N)}(\Delta) = \sum_{\theta_k\in\Delta}|\theta_k^{(N)}\rangle\langle\theta_k^{(N)}|$ converge weakly to $E(\Delta)$ on finitely supported vectors as $N\to\infty$. It is not projection valued, which is how it is compatible with Pauli's theorem [13, 14].*

Proof. (1) is immediate. For (2), with $|\chi\rangle = \sum_n c_n|n\rangle$, $\int_0^{2\pi}\frac{d\theta}{2\pi}\sum_{n,m} c_n^* c_m e^{i(n-m)\theta} = \sum_n |c_n|^2$; polarization gives $E(S^1) = I_C$. For (3), $U(\phi)|\theta) = |\theta+\phi)$ and the measure is translation invariant. For (4), $\langle\chi|E^{(N)}(\Delta)|\eta\rangle$ is a Riemann sum of $\int_\Delta \frac{d\theta}{2\pi}\langle\chi|\theta)(\theta|\eta\rangle$ once N exceeds the support. □

Remark 2. On a twisted fibre $\mathcal{H}_\kappa$ of Proposition 1 the same construction with $|\theta)_\kappa = \sum_{n\geq 0} e^{i(n+\kappa/2\pi)\theta}|n+\kappa/2\pi\rangle$ gives a covariant POVM with identical properties; the clock "number" then takes values in $\mathbb{N}_0 + \kappa/2\pi$.

IV. THE QUADRATURE CONJECTURE AND THE TRI-FOLD CONDITIONAL STRUCTURE

Definition 2 (Tri-fold quadrature). Let $|\Psi\rangle \in \mathcal{H}_C\otimes\mathcal{H}_S$, let $E(d\theta)$ be the clock POVM and let $|\Psi(\theta)\rangle = ((\theta|\otimes I_S)|\Psi\rangle$. Define

$$\text{raw clock marginal:} \qquad M_\Psi(\theta) = \tfrac{1}{2\pi}\|\Psi(\theta)\|^2, \tag{10}$$

$$\text{joint quadrature:} \qquad J_\Psi(\theta, F) = \tfrac{1}{2\pi}\langle\Psi(\theta)|F|\Psi(\theta)\rangle, \quad 0 \leq F \leq I_S, \tag{11}$$

$$\text{conditional state:} \qquad \rho_S(\theta) = \frac{|\Psi(\theta)\rangle\langle\Psi(\theta)|}{\|\Psi(\theta)\|^2}, \qquad P_\Psi(F|\theta) = \frac{J_\Psi(\theta,F)}{M_\Psi(\theta)} = \mathrm{Tr}[\rho_S(\theta)F]. \tag{12}$$

Where the circle is squared. Equation (10) is the precise step to which the title refers. The clock amplitudes $|\Psi(\theta)\rangle$ carry the $U(1)$ phases of Eq. (1), and no measurement can see their common part. The quadratic map $\|\cdot\|^2$ removes that common phase and converts the *relative* phases between clock levels, which are what $(\theta|$ reads out, into a nonnegative density over readings. Equations (11)–(12) apply the same squaring to the system, so that each reading carries a state. "Squaring" here means this quadratic (Born) map and nothing more; no relation to the classical compass-and-straightedge problem is implied.

Conjecture 1 (Hilbert-space quadrature conjecture)**.** *In a closed quantum universe there is no primitive external time parameter. Temporal order and dynamics are generated by the universal wave function through the quadratic map $\Psi \mapsto (M_\Psi, J_\Psi, \rho_S)$ over a covariant clock POVM; classical time $t = \theta/\omega$ (with winding number counted, Sec. II.D) is a derived semiclassical statistic of this structure.*

Theorem 1 (Conditional Schrödinger dynamics; cf. [4, 10, 11])**.** *Let $|\Psi\rangle$ satisfy Eq. (2) with $\mathcal{H}_C = H^2(S^1)$ and system energies $E_k = \hbar\omega n_k$, $n_k \in \mathbb{N}_0$. Then (i) $M_\Psi(\theta) = 1/2\pi$; (ii) $\rho_S(t) := \rho_S(\omega t) = |\psi(t)\rangle\langle\psi(t)|$ with $i\hbar\,\partial_t|\psi(t)\rangle = \widehat{H}_S|\psi(t)\rangle$.*

Proof. The constraint restricts $|\Psi\rangle = \sum_k \alpha_k|n_k\rangle|E_k\rangle$, $\sum_k |\alpha_k|^2 = 1$. Then $|\Psi(\theta)\rangle = \sum_k \alpha_k e^{-iE_k\theta/\hbar\omega}|E_k\rangle$, whose norm is 1, giving (i); differentiating with $\partial_\theta = \omega^{-1}\partial_t$ gives (ii). □

Theorem 1 has a simple reading in these terms. An exact constraint state squares the circle into a *flat* measure (i): every reading is equally weighted, so the universe has dynamics (ii) but no preferred moment. This is the PW answer to "frozen time": what is frozen is the marginal, not the conditional evolution. This is the standard PW result expressed with a covariant POVM; it holds equally on a twisted fibre (any $E_k \in \hbar\omega(\mathbb{N}_0 + \kappa/2\pi)$) and on the universal cover. It is verified numerically in Appendix A.

Proposition 4 (Peaked clocks from spectral windows)**.** *For $w \in \mathcal{S}(\mathbb{R})$ with $\int |w|^2 dt = 1$ and temporal width ΔT, let $|\Psi_w\rangle = \int dt\, w(t)|t\rangle \otimes e^{-i\widehat{H}_S t/\hbar}|\psi_0\rangle$. Then $\|\widehat{C}_{\rm univ}\Psi_w\| = \hbar\|w'\|_{L^2}$ ($= \hbar/2\Delta T$ for a Gaussian), $M_{\Psi_w}(t) = |w(t)|^2$, and the conditional state is $e^{-i\widehat{H}_S t/\hbar}|\psi_0\rangle$ on the support of w.*

Proof. $(-i\hbar\, d/dt + \widehat{H}_S)[w(t)e^{-i\widehat{H}_S t/\hbar}\psi_0] = -i\hbar w'(t)e^{-i\widehat{H}_S t/\hbar}\psi_0$; take norms. The rest follows from $\langle t|\Psi_w\rangle = w(t)e^{-i\widehat{H}_S t/\hbar}\psi_0$. □

Unitarity of the conditional evolution here holds by construction; the content of the proposition is the trade-off between a peaked marginal and an approximate constraint. Exact constraint states have uniform marginals (Theorem 1); peaked clocks require $\|\widehat{C}\Psi\| \neq 0$. This answers the familiar objection that a stationary universe cannot display a "present moment".

A. Wigner–Weyl quadrature

For $\mathcal{H}_S = L^2(\mathbb{R}^n)$ let $\psi(q,\theta) = ((\theta| \otimes \langle q|)|\Psi\rangle$ and define the conditional Wigner function

$$W_\Psi(q,p;\theta) = \frac{1}{(2\pi\hbar)^n}\int d^n y\, e^{-ip\cdot y/\hbar}\,\psi\big(q+\tfrac{y}{2},\theta\big)\,\psi^*\big(q-\tfrac{y}{2},\theta\big)\,. \tag{13}$$

Then $\int d^n q\, d^n p\, W_\Psi = 2\pi M_\Psi(\theta)$, and under Theorem 1, $\omega\,\partial_\theta W_\Psi = \{\{H_S, W_\Psi\}\}_{\rm Moyal} \to \{H_S, W_\Psi\}_{\rm PB}$ as $\hbar \to 0$. Geometrically, a single quadratic map acts on both factors. On the clock factor it produces the temporal weight $2\pi M_\Psi(\theta)$; on the system factor it produces a quasi-distribution on the classical phase space $\mathbb{R}^{2n}$, whose θ-flow becomes the Hamiltonian flow as $\hbar \to 0$. The circle supplies the ordering parameter, and the squared system amplitudes supply the arena in which classical motion along it takes place. Two cautions apply. First, $\mathbb{R}^{2n}$ is the phase space of the *system*; it is not the projective space $\mathbb{P}(\mathcal{H}) = S(\mathcal{H})/U(1)$ of Eq. (1), which is a Kähler manifold of real dimension $2(\dim\mathcal{H} - 1)$. Second, W_Ψ can be negative, so "distribution" is meant in the quasi-probability sense.

V. FINITE-CLOCK CORRECTIONS

Two physically different imperfections must be distinguished.

A remark on "clock energy spread". In an exact constraint state the clock's energy distribution is fixed by the system's: for $|\Psi\rangle = \sum_k \alpha_k |n_k\rangle|E_k\rangle$ it is $\{|\alpha_k|^2\}$. Dressing each clock level with a wave packet, $\sum_n g(n)|n+n_k\rangle$, contributes only a common factor $\hat{g}(\theta)$ to $|\Psi(\theta)\rangle$ and produces no dephasing. Dephasing requires either an unsharp *readout* or noise in, or interaction with, the clock [10].

A. Unsharp readout: a static effect

Proposition 5 (Unsharp covariant readout)**.** *Let the clock be read with the smeared covariant POVM* $E_\sigma(d\theta) = d\theta \int d\theta'\, G_\sigma(\theta-\theta')\, \frac{1}{2\pi}|\theta'\rangle\langle\theta'|$, *with* G_σ *a (wrapped) Gaussian of width* σ_θ. *For an exact constraint state,*

$$\rho_S^\sigma(\theta) = \int d\phi\, G_\sigma(\phi)\, \rho_S(\theta-\phi), \qquad \rho_{jk}^\sigma(\theta) = \rho_{jk}(\theta)\, e^{-\nu_{jk}^2\sigma_\theta^2/2} = \rho_{jk}(\theta)\, e^{-\omega_{jk}^2\sigma_t^2/2}, \tag{14}$$

with $\nu_{jk} = (E_j - E_k)/\hbar\omega$, $\omega_{jk} = (E_j - E_k)/\hbar$ *and* $\sigma_t = \sigma_\theta/\omega$ *(the wrapped-Gaussian correction is exponentially small for* $\sigma_\theta \ll \pi$*). The suppression is* independent of θ.

Proof. E_σ is a convex mixture of shifted copies of E and hence a covariant POVM. Since M_Ψ is uniform the smeared marginal is uniform, and the unnormalized conditional state is the G_σ-average of $|\Psi(\theta')\rangle\langle\Psi(\theta')|$. In the energy basis this multiplies ρ_{jk} by the characteristic function of G_σ at ν_{jk}. □

With a Fourier-limited resolution $\sigma_t \simeq \hbar/(2\Delta E_C)$ the factor is $\exp[-(E_j-E_k)^2/(8\Delta E_C^2)]$, which is negligible for any macroscopic clock (Sec. VI).

B. Diffusive clock error: the Lindblad form

Assumption 1 (Diffusive clock)**.** The timing error of the clock, relative to an ideal relational clock, is Gaussian with variance growing linearly, $\sigma_t^2(t) = \sigma_0^2 + T_C t$, with a phenomenological clock-diffusion time T_C.

Proposition 6 (Double-commutator master equation under Assumption 1)**.** *Let* $\rho_S^{\rm obs}(t) = \int d\tau\, G_{\sigma_t(t)}(\tau)\, U(t+\tau)\rho_S(0)U^\dagger(t+\tau)$. *Then (i) to second order in* σ_t, $\rho_S^{\rm obs}(t) = U(t)\big[\rho_S(0) - \frac{\sigma_t^2}{2\hbar^2}[\widehat{H}_S,[\widehat{H}_S,\rho_S(0)]]\big]U^\dagger(t) + \mathcal{O}(\sigma_t^4)$; *(ii) exactly,* $\rho_{jk}^{\rm obs}(t) = \rho_{jk}(0)e^{-i\omega_{jk}t-\omega_{jk}^2\sigma_t^2(t)/2}$; *(iii) for* $\sigma_0 = 0$, $\rho_S^{\rm obs}$ *obeys the GKSL equation*

$$\frac{d\rho_S}{dt} = -\frac{i}{\hbar}[\widehat{H}_S,\rho_S] - \frac{T_C}{2\hbar^2}[\widehat{H}_S,[\widehat{H}_S,\rho_S]], \qquad \Gamma_{jk} = \frac{T_C}{2\hbar^2}(E_j-E_k)^2, \tag{15}$$

with the single Hermitian jump operator $L = \sqrt{T_C}\,\widehat{H}_S/\hbar$.

Proof. (i) follows from expanding $U(\tau)\rho U^\dagger(\tau)$ to second order and using $\int G\tau = 0$, $\int G\tau^2 = \sigma_t^2$. (ii) is the Gaussian Fourier transform. Differentiating (ii) with $\sigma_t^2 = T_C t$ gives $\dot{\rho}_{jk} = (-i\omega_{jk} - \frac{T_C}{2}\omega_{jk}^2)\rho_{jk}$, which is Eq. (15) in the energy basis since $[\widehat{H}_S,[\widehat{H}_S,|E_j\rangle\langle E_k|]] = \hbar^2\omega_{jk}^2|E_j\rangle\langle E_k|$; finally $-\frac{1}{2}[H,[H,\rho]] = H\rho H - \frac{1}{2}\{H^2,\rho\}$. □

Remark 3. Equation (15) is the real-clock master equation of Egusquiza *et al.* [15]; it is also the structure of Milburn's intrinsic decoherence [16] and of the Gambini–Porto–Pullin mechanism [17] (which uses a different, non-linear growth of σ_t^2). Within standard quantum mechanics T_C is set by the measured phase noise of the clock. A common heuristic, $T_C \sim \hbar/\Delta E_C$, is only a dimensional estimate: it is not implied by Proposition 5, which gives a static effect, and it is not derived here. The conjecture acquires empirical content beyond standard quantum mechanics only if it supplies an independent law for T_C.

Figure 2 illustrates Theorem 1 and Propositions 5–6 computed from an explicit bipartite state.

VI. A LABORATORY TEST: CIRCUIT QED

Proposition 6 is the point at which the framework could, in principle, meet experiment, so we examine it immediately, before any relativistic extension. PW kinematics itself has been illustrated with entangled photons [25]. A test of finite-clock effects must specify (a) the null model, standard quantum mechanics with the measured clock noise, and (b) which energies enter the rate.

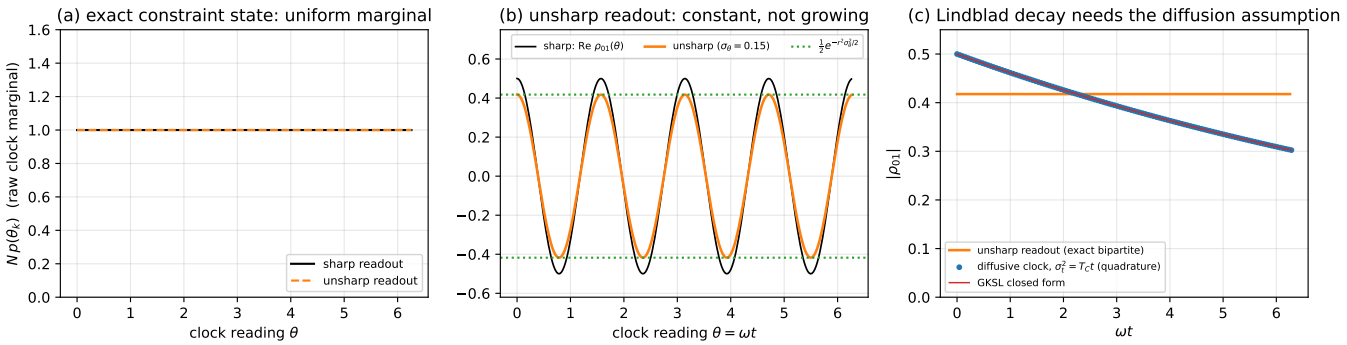


FIG. 2. Exact bipartite computation in $\mathbb{C}^{256} \otimes \mathbb{C}^2$ with system gap $r = 4$ clock quanta. (a) Raw clock marginal is uniform (Theorem 1). (b) Unsharp readout ($\sigma_\theta = 0.15$) suppresses the conditional coherence by the constant factor $e^{-r^2\sigma_\theta^2/2}$ (Proposition 5). (c) Exponential decay arises only under the diffusive Assumption 1; direct quadrature of the smearing channel coincides with the GKSL solution (Proposition 6).

Consider a transmon ($\omega_s/2\pi \approx 5$ GHz) dispersively coupled to a resonator ($\omega_c/2\pi \approx 6$ GHz) with $\widehat{H}_{\text{int}} = \hbar\chi a^\dagger a|1\rangle\langle 1|$, $\chi/2\pi \approx -1$ MHz, and use the resonator, prepared in a coherent state with $\Delta E_C = \hbar\omega_c\sqrt{\bar{n}}$, as the clock. Inserting the heuristic $T_C = \hbar/\Delta E_C$ into Eq. (15) gives $\Gamma = (\Delta E)^2/(2\hbar\Delta E_C)$. Table I and Fig. 3 compare two identifications of ΔE.

TABLE I. Model rate $\Gamma = (\Delta E)^2/(2\hbar\Delta E_C)$ for the circuit-QED parameters above (computed with `qtime.estimates`).

$\bar{n}$	Γ for $\Delta E = \hbar\chi$ [s^{-1}]	implied time	Γ for $\Delta E = \hbar\omega_s$ [s^{-1}]	implied T_2
10	166	6.0 ms	4.1×10^9	0.24 ns
10^2	52	19 ms	1.3×10^9	0.76 ns
10^3	17	60 ms	4.1×10^8	2.4 ns
10^4	5.2	0.19 s	1.3×10^8	7.6 ns

1. With $\Delta E = \hbar\omega_s$, the qubit's actual energy gap relative to the resonator clock, the model predicts $T_2 \lesssim 10$ ns, whereas transmons routinely reach $T_2 \sim 0.1$–1 ms. *The heuristic $T_C = \hbar/\Delta E_C$ with this identification is excluded.*

2. With $\Delta E = \hbar\chi$, rates of 5–170 s^{-1} lie one to three orders of magnitude below intrinsic qubit dephasing. Choosing χ amounts to working in a frame rotating at ω_s, which is defined by an external laboratory clock; this identification needs a justification that the framework does not currently provide.

3. The static effect of Proposition 5, with $\sigma_t = \hbar/2\Delta E_C$, is $1 - \mathcal{O}(10^{-10})$ and unobservable.

4. Heterodyne detection realizes a smeared covariant phase POVM (the phase marginal of the Husimi function), not the canonical Holevo POVM; the smearing must be included in the null model.

5. The dispersive Hamiltonian is valid only for $\bar{n} \ll n_{\text{crit}} = \Delta^2/4g^2 \approx \Delta/4\chi$, with $\Delta = \omega_c - \omega_s$. For $\Delta/2\pi \approx 1$ GHz and $\chi/2\pi \approx 1$ MHz, $n_{\text{crit}} \approx 250$, so the rows $\bar{n} = 10^3$–10^4 of Table I lie outside the model's own regime. Sweeping $\bar{n}$ over three decades inside it requires $\chi/2\pi \lesssim 2.5$ kHz, which lowers the χ-rate to 10^{-3}–10^{-5} s^{-1}.

Protocol. The intended discriminating experiment conditions the qubit coherence on a clock resonator whose phase noise is independently characterized. It holds the resonator linewidth κ, the temperature and the detuning fixed and sweeps the coherent-state amplitude, and hence $\Delta E_C \propto \sqrt{\bar{n}}$, at fixed χ inside the dispersive window. The signature of the heuristic is a residual dephasing rate scaling as $\bar{n}^{-1/2}$ after the null model has been subtracted. Two standard backgrounds have different $\bar{n}$ dependence. Measurement-induced dephasing grows linearly with $\bar{n}$ ($\propto \chi^2\bar{n}/\kappa$ for $\chi \ll \kappa$), and thermal-photon dephasing is set by the thermal occupation, independent of the drive. The opposite, decreasing trend is therefore a useful handle. Items 1 and 5 nevertheless bound what the protocol can reach. The ω_s identification is already excluded, and the χ identification gives rates far below intrinsic qubit dephasing once the dispersive constraint is respected. A null result would therefore constrain the heuristic $T_C \sim \hbar/\Delta E_C$, not the

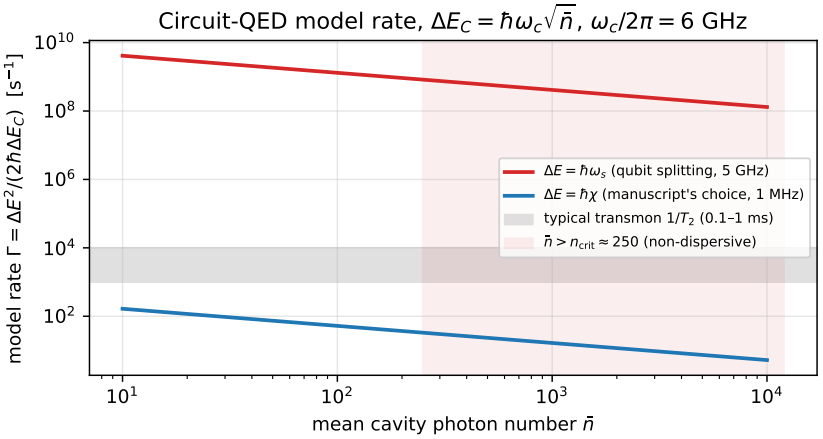


FIG. 3. Model rate of Sec. VI versus mean photon number for the two identifications of the system energy gap, compared with typical transmon dephasing rates. Points with $\bar{n} \gtrsim n_{\rm crit} \approx 250$ lie outside the dispersive regime (item 5).

conjecture as such. The winding-marginalization effect (Proposition 2) is a robust standard-quantum-mechanics prediction for cyclic clocks that any implementation should reproduce, but it does not discriminate the conjecture from standard quantum mechanics.

VII. CONSISTENCY WITH THERMAL TIME

Proposition 7 (Modular flow for Gibbs Schmidt spectra)**.** *Let* $|\Psi\rangle = \sum_k \alpha_k |n_k\rangle|E_k\rangle$ *with all* $\alpha_k \neq 0$*, and let* $\mathcal{A}_S = I_C \otimes B(\mathcal{H}_S)$ *act on the closure of* $\mathcal{A}_S|\Psi\rangle$*. The modular operator is*

$$\Delta_\Psi = \rho_C^{-1} \otimes \rho_S, \qquad \sigma_s^\Psi(I \otimes A) = \Delta_\Psi^{is}(I \otimes A)\Delta_\Psi^{-is} = I \otimes \rho_S^{is} A \rho_S^{-is}. \tag{16}$$

If $|\alpha_k|^2 = Z^{-1}e^{-\beta E_k}$*, then* $\sigma_s^\Psi(A) = e^{i\widehat{H}_S t/\hbar} A e^{-i\widehat{H}_S t/\hbar}$ *with* $t = -\hbar\beta s$*.*

Proof. For the Tomita operator $S(I\otimes A)|\Psi\rangle = (I\otimes A^\dagger)|\Psi\rangle$ a direct computation in the Schmidt basis gives $S^*S = \rho_C^{-1}\otimes \rho_S$ (we have also checked this numerically by constructing S from its definition; Appendix A). With $\rho_S = Z^{-1}e^{-\beta\widehat{H}_S}$, $\rho_S^{is} A \rho_S^{-is} = e^{-is\beta\widehat{H}_S} A e^{is\beta\widehat{H}_S}$, which equals the Heisenberg evolution at $t = -\hbar\beta s$. □

This is the KMS content of the thermal-time hypothesis [6] restricted to the special case of a Gibbs entanglement spectrum. For non-Gibbs spectra modular flow and conditional (quadrature) time differ.

VIII. LOCAL RELATIVISTIC REGIME: FROM FLAT SPACETIME TO THE HORIZON

Sections VIII and IX subject the conjecture to a progressive stress test, ordered by regime. Here the clock still lives in a fixed background: first flat spacetime (Klein–Gordon, Dirac, proper time), then a static gravitational field, and finally the near-horizon limit. Section IX removes the background. Each result is either established (and attributed) or explicitly marked as heuristic.

A. Klein–Gordon constraint and positive frequency

With $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, $\widehat{P}_0 = \widehat{P}_t/c$ and $\widehat{M} = mI + \widehat{H}_{\rm int}/c^2$, the mass-shell constraint factorizes as

$$\left(\widehat{P}_\mu \widehat{P}^\mu - \widehat{M}^2 c^2\right) = c^{-2}\,\widehat{C}_+\widehat{C}_-, \qquad \widehat{C}_\pm = \widehat{P}_t \pm \widehat{H}_{\rm rel}, \quad \widehat{H}_{\rm rel} = \sqrt{c^2\widehat{\mathbf{P}}^2 + \widehat{M}^2c^4} > 0. \tag{17}$$

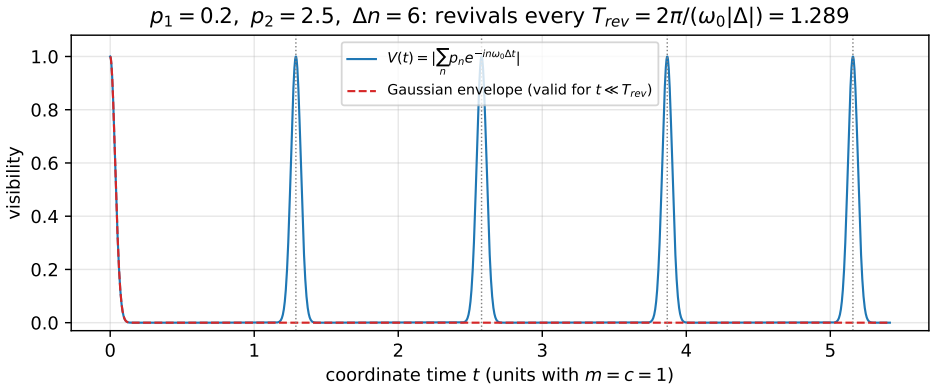


FIG. 4. Relativistic clock interferometer: exact visibility (solid) with full revivals, and the short-time Gaussian envelope (dashed).

Proposition 8 (Semi-bounded clocks select positive frequency)**.** *Restrict the universal-cover clock to the Hardy space of the line, i.e. to clock energy* $\widehat{H}_C := -\widehat{P}_t \geq 0$ *(the line analogue of Remark 1). Then* $\ker \widehat{C}_-$*, on which* $\widehat{H}_C = -\widehat{H}_{\rm rel} < 0$*, is annihilated, and conditional states on* $\ker \widehat{C}_+$ *obey* $i\hbar\partial_t\psi = \widehat{H}_{\rm rel}\psi$ *with nonnegative density* $|\langle \mathbf{x}|\psi(t)\rangle|^2$*.*

This is the standard positive-frequency (Newton–Wigner) restriction [26], obtained here by the choice of a semi-bounded clock; the associated localization is not Lorentz covariant and spreads acausally [27]. Working on the line rather than the circle avoids the commensurability issue for the continuous spectrum of $\widehat{H}_{\rm rel}$ (Corollary 1).

B. Dirac constraint and Zitterbewegung

For $\mathcal{H}_S = L^2(\mathbb{R}^3) \otimes \mathbb{C}^4$ the constraint $\widehat{P}_t + \widehat{H}_D$, $\widehat{H}_D = c\boldsymbol{\alpha}\cdot\widehat{\mathbf{P}} + \beta mc^2$, has energies $\pm E_{\mathbf{p}}$. A semi-bounded clock as in Proposition 8 removes the negative-energy branch altogether (the Foldy–Wouthuysen sector [28]), so no Zitterbewegung remains. Only for a two-sided (universal-cover) clock subject to Assumption 1 do the interference terms between branches decay, at rate $\Gamma_{\rm Zitt} = T_C(2E_{\mathbf{p}})^2/2\hbar^2 \geq 2m^2c^4T_C/\hbar^2$; with the heuristic $T_C \sim \hbar/\Delta E_C$, $\Gamma_{\rm Zitt} \gtrsim 2m^2c^4/(\hbar\Delta E_C)$.

C. Internal proper-time quadrature

For an internal Hardy clock $\widehat{H}_{\rm int} = \hbar\omega_0\widehat{N}_{\rm int}$ carried by a particle of mass m,

$$\widehat{H}_{\rm rel} = \widehat{E}_{\mathbf{P}} \otimes I + \widehat{\gamma}(\widehat{\mathbf{P}})^{-1} \otimes \hbar\omega_0\widehat{N}_{\rm int} + \mathcal{O}\Big(\widehat{H}_{\rm int}^2/mc^2\Big), \qquad \widehat{\gamma} = \sqrt{1 + \widehat{\mathbf{P}}^2/m^2c^2}, \tag{18}$$

so the internal-clock reading conditioned on laboratory time satisfies $\langle\tau(t)\rangle = \langle\widehat{\gamma}^{-1}\rangle t$. A momentum superposition $c_1|\mathbf{p}_1\rangle + c_2|\mathbf{p}_2\rangle$ with the internal clock in a superposition of two levels n, m shows the quantum twin-paradox modulation $|c_1|^4 + |c_2|^4 + 2|c_1|^2|c_2|^2\cos[(n-m)\omega_0(\gamma_1^{-1} - \gamma_2^{-1})t]$ [21]. For a broad internal distribution p_n the centre-of-mass visibility is $V(t) = |\sum_n p_n e^{-in\omega_0\Delta t}|$ with $\Delta = \gamma_1^{-1} - \gamma_2^{-1}$. Because the internal spectrum is equally spaced, V revives exactly with period $2\pi/(\omega_0|\Delta|)$; the Gaussian envelope $e^{-\Delta n^2\omega_0^2\Delta^2 t^2/2}$ describes only the first collapse (Fig. 4).

D. Static spacetimes

For $ds^2 = N(\mathbf{x})^2c^2dt^2 - h_{ij}dx^idx^j$ (Schwarzschild exterior: $N = \sqrt{1 - r_s/r}$, $r_s = 2GM/c^2$), conditioning on an asymptotic clock gives the Killing Hamiltonian

$$\widehat{H}_{\rm grav} = N(\widehat{\mathbf{X}})^{1/2}\sqrt{c^2\widehat{\mathbf{P}}_\Sigma^2 + \widehat{M}^2c^4}\; N(\widehat{\mathbf{X}})^{1/2}. \tag{19}$$

Proposition 9 (Gravitational time dilation and clock decoherence)**.** *In the quasi-static regime* $\widehat{H}_{\rm grav} \approx \widehat{H}_{\rm cm} \otimes I + N(\widehat{\mathbf{X}}) \otimes \hbar\omega_0 \widehat{N}_{\rm int}$*. Then: (i) for a probe at radius* r*, the conditional internal-clock distribution is translated,* $Q(\tau|t,r) = Q(\tau - N(r)t\,|\,0,r)$*; (ii) for a superposition of two shells with* $\Delta N_{12} = N(r_2) - N(r_1)$ *and a Gaussian internal energy spread* $\Delta E_{\rm int} = \hbar\omega_0 \Delta n$*, the centre-of-mass visibility is* $\mathcal{V}(t) = \exp[-(\Delta E_{\rm int}\Delta N_{12} t)^2/2\hbar^2]$*, i.e.* $\tau_{\rm grav} = \sqrt{2}\hbar/(\Delta E_{\rm int}|\Delta N_{12}|)$ *[18, 19]; (iii)* (model-dependent) *if a clock at* r_C *with local energy spread* $\Delta E_C^{\rm loc}$ *times a system at* r_S *under Assumption 1 with* $T_C \sim \hbar/\Delta E_C$ *evaluated in Killing time, the coordinate-time rate is*

$$\Gamma_{jk}^{\rm grav} = \frac{N(r_S)^2}{N(r_C)} \, \frac{(E_j^{\rm loc} - E_k^{\rm loc})^2}{2\hbar \Delta E_C^{\rm loc}}. \tag{20}$$

Proof. (i) The level $|n\rangle_{\rm int}$ acquires phase $e^{-in\omega_0 N(r)t}$, shifting the argument of the covariant POVM by $N(r)t$. (ii) Tracing out the internal clock gives $\langle r_1|\rho_{\rm cm}|r_2\rangle \propto \sum_n |c_n|^2 e^{in\omega_0 \Delta N_{12} t}$, the characteristic function of a Gaussian (valid for t much shorter than the revival time). (iii) Killing-time energy gaps are $N(r_S)\Delta E^{\rm loc}$ and the clock's Killing energy spread is $N(r_C)\Delta E_C^{\rm loc}$; substitute into Eq. (15). □

For co-located clock and system, transforming Eq. (20) to local proper time removes the redshift factors, as the equivalence principle requires. Near the horizon, in terms of the proper distance ρ from r_s, $N(r) \simeq \kappa\rho/c^2$ with $\kappa = c^4/4GM$, so $\Gamma^{\rm grav} \propto N(r_C)^{-1} \propto \rho^{-1}$, equivalently $\propto (r_C - r_s)^{-1/2}$ in the coordinate distance. This divergence reflects the vanishing Killing-energy spread of a near-horizon clock; it is not a thermal effect.

Remark 4 (Heuristic optimal bandwidth). If one simply adds the rate $a/\Delta E_C$, $a = (E_j - E_k)^2/2\hbar$, to the inverse Gaussian time $b\,\Delta E_C$, $b = |\Delta N_{12}|/\sqrt{2}\hbar$, of (ii), the sum is minimized at $\Delta E_C^* = |E_j - E_k|/(2^{1/4}|\Delta N_{12}|^{1/2})$ with minimum $2\sqrt{ab} = 2^{1/4}|E_j - E_k|\,|\Delta N_{12}|^{1/2}/\hbar$. This is a scaling guide only: it combines an exponential with a Gaussian decay law and two different coherences. In the laboratory, $\Delta N_{12} \approx g\Delta x/c^2 \approx 1.1 \times 10^{-16}$ for $\Delta x = 1$ m, so $\Delta E_C^* \approx 8 \times 10^7\,|E_j - E_k|$.

E. Near-horizon consistency with Bisognano–Wichmann and Sewell

Near $r = r_s$ the metric is approximately Rindler in the proper distance ρ, $ds^2 \simeq (\kappa\rho/c^2)^2 c^2 dt^2 - d\rho^2 - r_s^2 d\Omega^2$. By the Bisognano–Wichmann theorem [31, 32] and Sewell's extension to Kruskal spacetime [33], the modular group of the (Minkowski, respectively Hartle–Hawking–Israel) vacuum restricted to the wedge is the boost flow, $\Delta^{is} = e^{-2\pi i s K}$, where the dimensionless boost generator satisfies $K = c\widehat{H}_{\rm Killing}/(\hbar\kappa)$. Hence σ_s is Killing-time evolution with $t = -2\pi c s/\kappa$. Identifying this with the Gibbs relation $t = -\hbar\beta s$ of Proposition 7 gives

$$k_B T_H = \frac{\hbar\kappa}{2\pi c} = \frac{\hbar c^3}{8\pi G M}, \tag{21}$$

the Hawking–Unruh temperature [34, 35] (6.17×10^{-8} K for one solar mass). The temperature is supplied by the Bisognano–Wichmann/Sewell theorem; the quadrature framework adds only the identification of the modular parameter with a clock reading. Moreover, the wedge algebra is of type III_1, so expressions such as $\rho_{\rm wedge}$ and $K = -\ln \rho_{\rm wedge}$ are formal; Proposition 7, proved for type I algebras, applies only by analogy.

IX. COSMOLOGICAL AND BOUNDARY REGIME

The last rung of the ladder removes the external background altogether: the clock is now a degree of freedom of the universe itself, and there is no asymptotic observer whose time could be borrowed. Only Proposition 10 is a result; the rest of this section is explicitly heuristic.

A. Minisuperspace cosmology with a scalar-field clock

For a flat FRW universe with scalar field φ ($\hbar = c = 1$), the Hamiltonian constraint is $-p_a^2/(2m_0 a) + p_\varphi^2/(2a^3) + a^3 V(\varphi) \approx 0$, with $m_0 \propto G^{-1}$ (its value depends on the fiducial-volume convention). With the ordering $a^{-p}\partial_a a^p \partial_a$,

$$\left[\frac{1}{2m_0 a}\, a^{-p}\partial_a(a^p\partial_a) - \frac{1}{2a^3}\partial_\varphi^2 + a^3 V(\varphi)\right]\Psi(a,\varphi) = 0 \iff -\partial_\varphi^2 \Psi = \widehat{\Theta}(\varphi)\Psi, \quad \widehat{\Theta} = -\frac{a^2}{m_0} a^{-p}\partial_a(a^p\partial_a) - 2a^6 V(\varphi). \tag{22}$$

Proposition 10 (Exact case: massless scalar clock)**.** *For $V = 0$ and $p = 1$, $\widehat{\Theta} = -m_0^{-1}\partial_x^2$ with $x = \ln a$ is positive and self-adjoint on $L^2(\mathbb{R}, dx)$. Positive-frequency physical states satisfy $i\partial_\varphi \psi(x|\varphi) = \sqrt{\widehat{\Theta}}\,\psi(x|\varphi)$ with respect to the scalar-field clock, and the conditional scale-factor distribution $|\psi(x|\varphi)|^2$ evolves unitarily in φ.*

This is the Wheeler–DeWitt model underlying loop quantum cosmology [29], written in quadrature language; the scalar clock lives on the line with unbounded momentum, i.e. on the universal cover of Sec. II. For $V(\varphi) \neq 0$, $\widehat{\Theta}(\varphi)$ depends on the clock and is not positive for $V > 0$ at large a, so a first-order equation holds only approximately, in adiabatic regions where $\widehat{\Theta} > 0$.

B. Euclidean clocks and the no-boundary state (heuristic)

Conjecture 2 (Euclidean quadrature reading of the no-boundary proposal)**.** *The compact Euclidean "time" direction of no-boundary geometries [30] can be interpreted as the phase circle S^1_E of a Euclidean clock sector, with the no-boundary wave function arising as a conditional amplitude of a clock–geometry state, and with the transition to Lorentzian evolution corresponding to unwrapping S^1_E onto the universal cover of Sec. II.*

We stress that this is a programme, not a result. The Hartle–Hawking wave function depends on boundary data (h_{ij}, φ), with Euclidean time integrated over, and the Euclidean action is not a generator of clock translations. A concrete realization would have to identify the clock sector, its Hilbert space (twisted or Hardy) and the map from path-integral data to clock amplitudes.

C. Open directions in this regime

Two further cosmological applications were proposed in earlier drafts, and both are still open. (i) *Clock-induced Hubble-rate variance.* A finite scalar-field clock should make the conditional expansion rate fluctuate, but no such variance follows from Eq. (22) without an additional ansatz; a derivation would need the analogue of Assumption 1 for φ. (ii) *Back-reaction on inflationary perturbations.* A naive Gaussian clock-smearing factor evaluated at horizon crossing is independent of k, so it rescales the power spectrum without tilting it. An observable effect would have to come from a k-dependent clock coupling, which has not been constructed.

X. DISCUSSION AND OPEN PROBLEMS

The Hilbert-space quadrature conjecture organizes relational quantum time around one picture: a circular phase fibre, made observable by entanglement and turned into a nonnegative temporal measure by the quadratic map. Its mathematical backbone is the PW construction with covariant clock POVMs. Within that backbone we have (i) shown how the line and the circle are related, via twisted sectors and winding marginalization; (ii) separated static readout smearing from diffusive clock error; and (iii) shown that the circuit-QED signature depends on an energy identification that the framework must still justify, and that the dispersive regime limits the photon-number sweep. Applied rung by rung (flat spacetime, static gravity, the horizon, cosmology), the framework reproduces the established results (positive-frequency selection, proper-time and gravitational dephasing, the Bisognano–Wichmann temperature) in a single language. Its genuinely new content shrinks as the regime becomes more extreme, ending at Conjecture 2.

Open problems include: a principled law for the clock-diffusion time T_C (without one the conjecture makes no predictions beyond standard quantum mechanics); the two cosmological directions of Sec. IX; a non-Abelian generalization with gauge-covariant constraints; and a concrete realization of Conjecture 2.

Appendix A: Numerical model

The accompanying Python package `qtime` builds explicit bipartite states $|\Psi\rangle \in \mathbb{C}^N \otimes \mathbb{C}^d$ that solve Eq. (2) for integer system levels. It constructs sharp (Pegg–Barnett) and smeared covariant clock POVMs and computes $p(\theta_k)$ and $\rho_S(\theta_k)$ by partial traces. An automated test suite checks: completeness, positivity and covariance of the POVMs; the uniform marginal and exact conditional evolution of Theorem 1; the absence of solutions off the clock spectrum; the θ-independent suppression of Proposition 5; agreement of direct quadrature, closed form and ODE integration for Proposition 6; the bound of Proposition 2; the corrected Szegő kernel (9); the modular operator (16), obtained

by constructing the Tomita operator from its definition; interferometer revivals; the near-horizon scalings; and the numbers in Table I. Figures 1–4 are generated by `scripts/make_figures.py`.

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